

KARAN PRABHAT

Aspiring AI/ML Engineer | Data Science | Full-Stack Developer

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Summary

Aspiring AI/ML engineer with strong foundations in machine learning, deep learning, and full-stack development. Proven ability to build end-to-end data science pipelines and deploy production-ready applications with measurable impact. Passionate about solving real-world problems through intelligent systems and targeting high-impact engineering roles at top-tier tech companies.

Education

- B.Tech — Computer Science & Engineering (AI & ML)** Aug 2023 – Aug 2027
Sister Nivedita University, Kolkata, West Bengal
- I.Sc — Intermediate of Science (Mathematics)** Apr 2020 – Apr 2022
Marwari College Ranchi, Jharkhand

Technical Skills

Programming Languages: Python, C, C++, JavaScript

AI/ML & Deep Learning: Machine Learning, ANN, Deep Learning, TensorFlow, Keras, Scikit-learn, SMOTE, imbalanced-learn

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, EDA, Jupyter Notebook, Colab

Web Development: HTML, CSS, React, Streamlit, Frontend, Backend Basics, REST APIs

Tools & Platforms: Git, GitHub, VS Code, Figma, Streamlit Cloud

Core CS Subjects: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Projects

IoT Network Security Analysis & ML-Based Attack Detection *Python, Pandas, NumPy, Scikit-learn, Matplotlib*

- Processed 211,043 IoT network flow records; built a full preprocessing pipeline (deduplication, one-hot encoding) transforming 44 raw columns into 51 engineered features across 190,474 clean records.
- Trained and benchmarked Logistic Regression, Decision Tree, and Random Forest classifiers; **Random Forest achieved 99.82% accuracy, F1 = 0.9988, Recall = 0.9990** (only 29 missed attacks out of 29,687).
- Derived 8 EDA-driven security insights — TCP carrying 89.2% of attacks, Port 4444 as backdoor indicator, and S0 state as the strongest port-scan signal.

FraudNet AI — Credit Card Fraud Detection *Python, TensorFlow/Keras, Scikit-learn, imbalanced-learn*

- Built an MLP (64→32→16→1) on 284,807 transactions (Kaggle ULB, 0.17% fraud rate); applied SMOTE exclusively on training data to prevent leakage, balancing distribution to 50% fraud.
- Achieved **AUC-ROC of 0.97–0.99** and **Recall of 0.85–0.93** on the original imbalanced test set; applied Dropout (0.3/0.3/0.2) and Early Stopping (patience=5) to prevent overfitting.

Canteen Menu Optimizer *Python, Scikit-learn, Streamlit | Deployed on Streamlit Cloud*

- Built a classification ML pipeline to predict dietary preferences; deployed an interactive Streamlit app achieving **82.6% accuracy, 56.4% estimated ROI improvement, and ~30% food waste reduction.**

Experience & Training

Full-Stack Web Developer — Training Program May 2024 – Aug 2024
Pantech Solutions, Chennai (Online)

- Built full-stack web application clones; implemented frontend–backend integration patterns, strengthening end-to-end development, debugging, and code-structuring skills.

InnovateBank Digital Transformation — Product Design & Implementation Nov 2025 – Jan 2026
XPMC XPro Program | Employability.life & Federation University Australia

- Redesigned a personal loan journey using the Double Diamond framework; conducted user research (surveys & interviews), designed Figma wireframes, and built a stage-wise Customer Communication Logic Framework.
- Delivered a phased roadmap (Pilot → Limited Release → Full Launch) with KPIs and a Complexity–Impact Matrix; contributed to agile sprint execution in a cross-functional team.

Achievements & Certifications

- Smart India Hackathon (Internal Round):** Ranked in top 35% among 120 teams (600+ participants) with an AI-based Traffic Management System project.
- XPMC XPro Certification** — Employability.life & Federation University Australia (2026): Product Thinking, UX Design, Roadmapping, and Agile Collaboration.

Certifications: Introduction to Generative AI — Google Cloud | Introduction to Artificial Intelligence (AI) — Coursera | Design Thinking for Innovation — Coursera | Internship in Full-Stack Web Development — Pantech Solutions